Computing Fundamentals
EC-110

Lecture # 01

About Me

- Professional Career
  - Dean HITEC University
  - Chairman Computer Engineering HITEC University
  - Ex Associate HoD Computer Engineering, College of E&ME, NUST
  - GM Corporate Affairs EME ICE
  - Consultant Punjab Higher Education Commission
  - Consultant Naya Pakistan Housing and Development Authority
  - Ex Consultant PM Laptop Scheme
  - Provided Consultancies to MO Railways, MOPS
  - Board of Directors of 3 IT firms
  - More than 110 International Publications (Cum IF = 60+)
  - Working with 4 International Collaborating Universities (US, UK, Canada)
- Professional Education
  - PhD Computer Engineering, UK (Fully Funded Scholarship)
  - MS Satellite Communication, UK (Fully Funded Scholarship)
  - BS Computer Engineering, NUST (Partially Funded Scholarship)

Policies

- The Lecture will always start on time, otherwise the changed schedule/time table will be communicated with you in advance.
- The quiz once taken will not be re-taken. There will be no makeup quiz.
- Assignments can be submitted before or on mentioned time.
  - If submitted after 24 hours of the due time, half marks shall be granted.
  - After that zero marks will be given.

Honesty is the best Policy!

- **Penalty for Cheating Cases:**
  - First Offense: Zero in assignment/lab/project for all involved.
  - Further Offenses: Zero in assignment/lab/project and -2 from internals/pre-finals theory marks for all involved.

Course Division

- Course Code EC-110
- Credit Hours (2+1)
- Theory (67%) + Laboratory (33%)

- Theory
  - Assignments (5-10)
  - Quizzes (5-10)
  - Sessional (10+10 = 20)
  - Final (30-40)

Course Outline

- Introduction to Computer
- Introduction to Number Systems
- Hardware Devices(i.e. I/O devices)
- Memory & Storage
- Boolean Algebra & Logic Gates
- Processor
- Viruses and Anti-Viruses
- Algorithms and Flowcharts
- Basic Programming Skills

Text and Material

- **Text Book**
  - Introduction to Computer By **Peter Norton 6th Edition**
  - Programming with C++ Object Oriented Programming By **Aikman Series**
  - Lecture Notes
  - Slides

- **Reference Material**
  - Wikipedia
  - Google Search

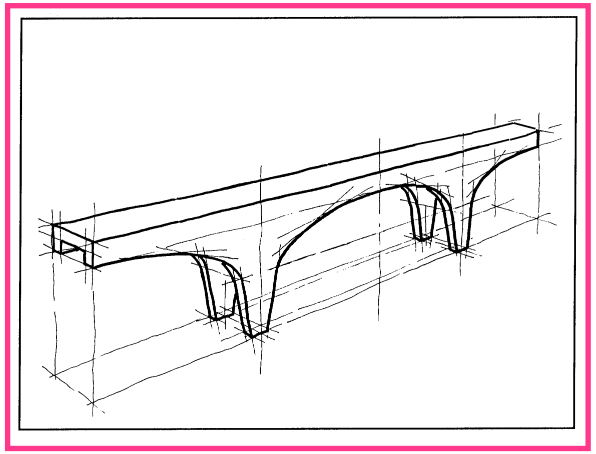
What is Engineering?

- Engineering is the profession in which knowledge of the mathematical and natural sciences is gained by study, experience, and practice is applied with judgment to develop ways to utilize, economically, the materials and forces of nature for the benefit of mankind.
- An *engineer* is a person who uses scientific knowledge to solve practical problems.
- Engineers create new things and make old things better.

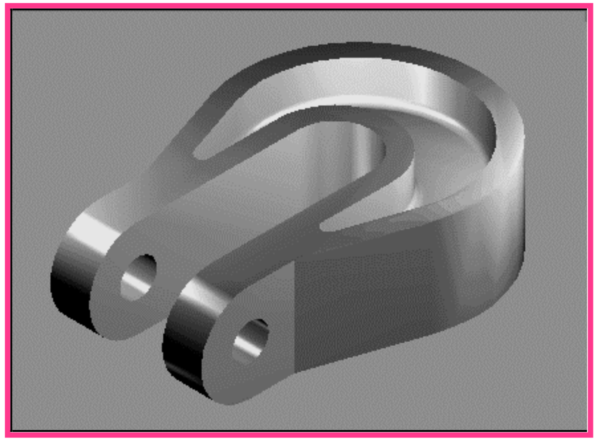
Basic Engineering Design Process

- Problem Identification: Get with Customer.
- Conceptual Design: Ideas, Sketches and Solution Lists.
- Refinement: Computer Modeling, Data Base Development.
- Testing: Analysis and Simulation of All Design Aspects.
- Prototyping: Visualizing and Improving the Design.
- Communication: Engineering Drawings, Specifications.
- Production: Final Design, Manufacturing, Distribution.

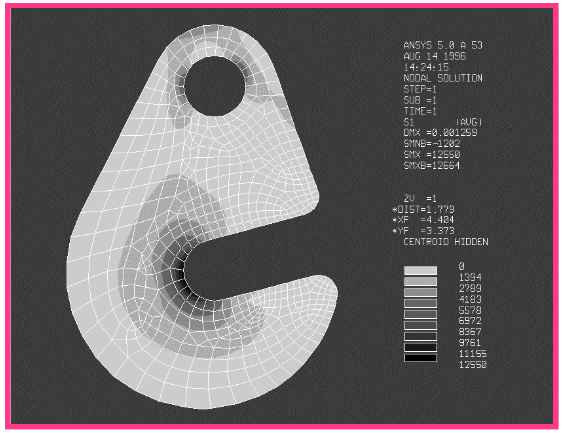
Design Graphics



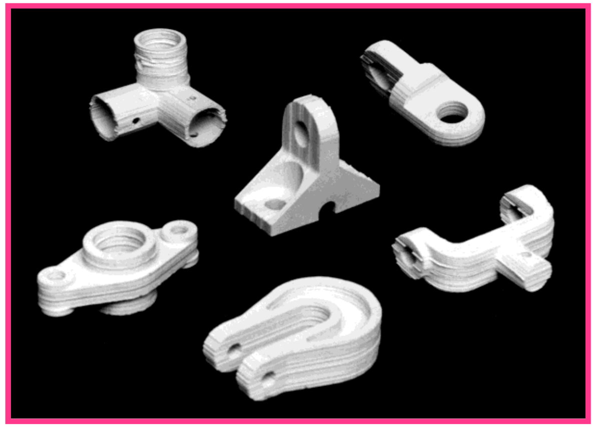
Sketching



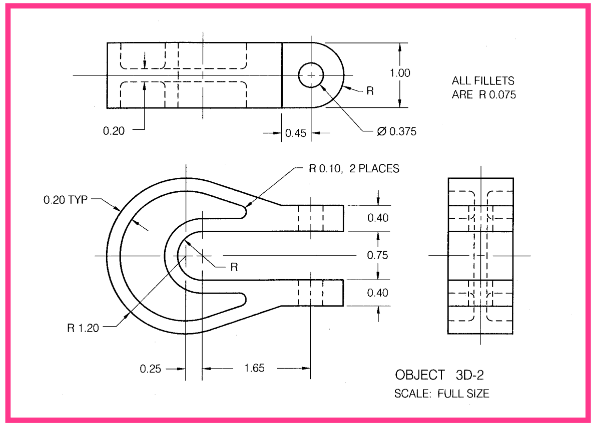
3-D Modeling



Analysis



Prototyping



2-D Drawing

Chapter # 01

Introducing Computer Systems

Chapter #1A

EXPLORING COMPUTERS AND THEIR USES

Overview

- Computer Defined
- Computers for Individual Users
- Computers for Organizations
- Computers in Society

The Computer Defined

- Electronic device
- Converts data into information
- Any computer regardless of its type, is controlled by instructions, which tell it what to do.
- Digital Computers
- Analogue Computers
- Can be categorized on different basis

- Modern computers are digital
  - They are so called because they work by the numbers.
  - Digital systems represent data as one distinct value or the other.
  - Break all types of information into tiny units, works on them and again combine them to represent information.
  - Work in a strict manner by processing these units individually and in organized way
- Older computers were analog
  - A range of values made data.
  - Somewhat more flexible but not necessarily more precise and reliable.

Computers For Individual Use

- Some computers are meant to be used by one person at a time.
- This category includes:
    - Desktop computers
    - Workstations
    - Notebook computers
    - Tablet computers
    - Handheld computers
    - Smart phones
- PCs are also called microcomputers for their size.
- Although used by individuals but can be part of networks.

Computers For Individual Use

- **Desktop computers**
  - The most common type of computer
  - Sits on the desk or floor
  - Performs a variety of tasks
  - Has a system unit
  - Might be horizontal or vertical
- **Workstations**
  - Specialized computers
  - Optimized for science or graphics
  - More powerful than a desktop

Computers For Individual Use

- **Notebook computers**
  - Small portable computers
  - Weighs between 3 and 8 pounds
  - About 8 ½ by 11 inches
  - Typically as powerful as a desktop
  - Can include a docking station

Computers For Individual Use

- **Tablet computers**
  - Newest development in portable computers
  - Input is through a pen called stylus or digital pen
  - Run specialized versions of office products

Computers For Individual Use

- **Handheld computers**
  - Very small computers
  - Personal Digital Assistants (PDA)
  - Note taking or contact management
  - Data can synchronize with a desktop
- **Smart phones**
  - Hybrid of cell phone and PDA
  - Web surfing, e-mail access

Computers For Organizations

- Some computers handle the needs for many people at the same time.
- Such systems normally lie at the heart of organizations network.
- Some are generic and some are special purpose.
- People generally access them through terminals.
- They include:
    - Network Servers
    - Mainframe Computers
    - Minicomputers
    - Supercomputers

Computers For Organizations

- **Network servers**
  - Centralized computer
  - All other computers connect
  - Provides access to network resources
  - Multiple servers are called clusters or server farms
  - Often simply a powerful desktop
  - Or might be mounted on in large racks or reduced to small units called Blades
  - Different servers may have different purposes

Computers For Organizations

- **Mainframes**
  - Used in large organizations
  - Handle thousands of users
  - Users access through a terminal
  - Two types of terminals
  - *Dumb terminal*→ only for input and output data
  - *Intelligent terminal*→ performs some processing but usually doesn't have any storage

Computers For Organizations

- **Minicomputers**
  - Called midrange computers
  - Power between mainframe and desktop
  - Handle hundreds of users
  - Used in smaller organizations
  - Users access through a terminal

Computers For Organizations

- **Supercomputers**
  - The most powerful computers made
  - Handle large and complex calculations
  - Process trillions of operations per second
  - Found in research organizations

Computers In Society

- More impact than any other invention
  - Changed work and leisure activities
  - Used by all demographic groups
- Computers are important because:
  - Provide information to users
  - Information is critical to our society
  - Managing information is difficult

Computers In Society

- **Computers at home**
  - Many homes have multiple computers
  - Computers are used for
    - Business
    - Entertainment
    - Communication
    - Education

Computers In Society

- **Computers in education**
  - Computer literacy required at all levels
- **Computers in small business**
  - Makes businesses more profitable
  - Allows owners to manage
- **Computers in industry**
  - Computers are used to design products
  - Assembly lines are automated

Computers In Society

- **Computers in government**
  - Necessary to track data for population
    - Police officers
    - Tax calculation and collection
  - Governments were the first computer users

Computers In Society

- **Computers in health care**
  - Revolutionized health care
  - New treatments possible
  - Scheduling of patients has improved
  - Delivery of medicine is safer

Evolution of the Computer

Evolution of the Computer

“***Computer generation” is a term related to the evolution and adaptation of technology and computing.***

- Reducing the size of processors.
- Increasing capacity and speed.

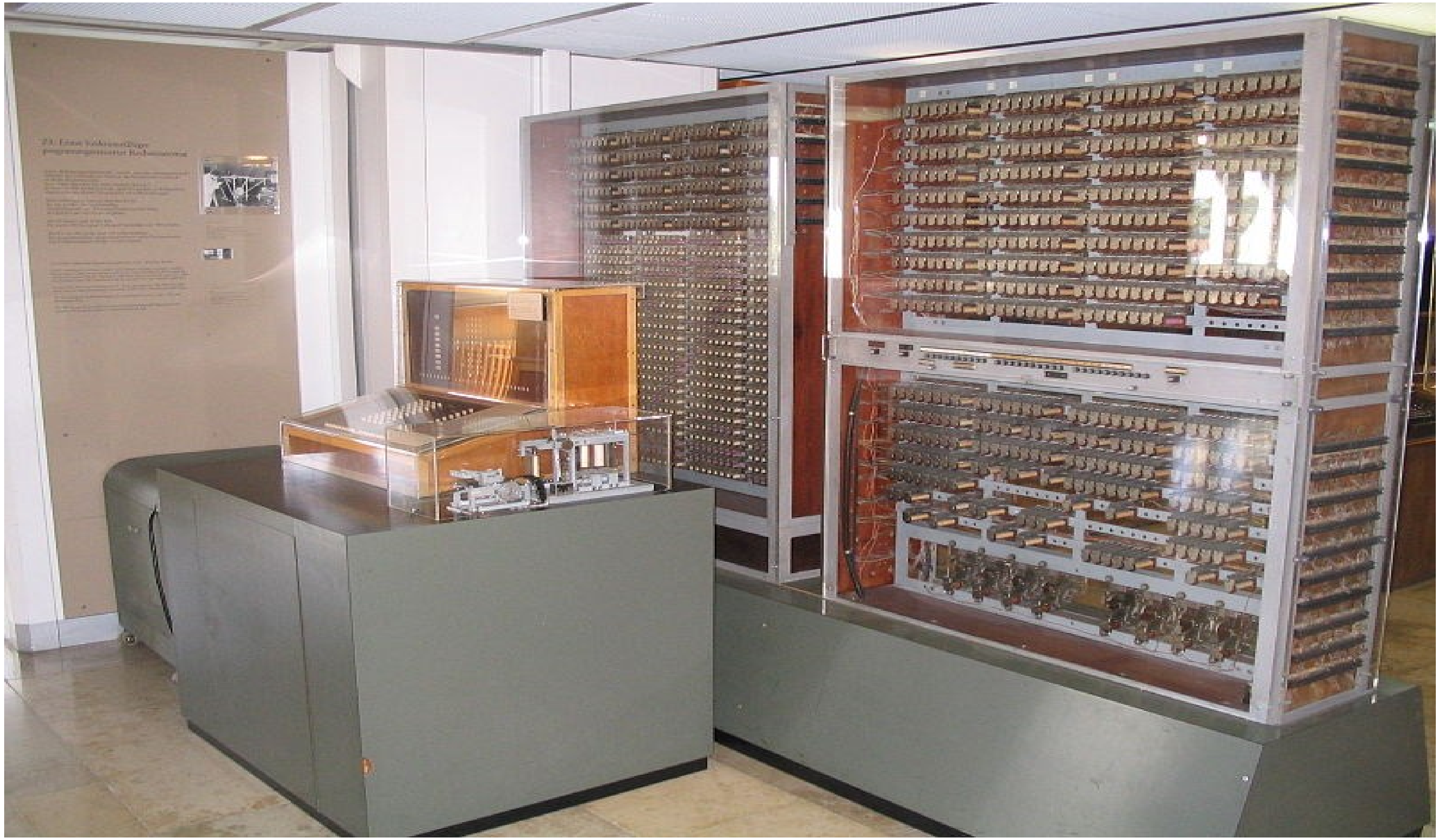
Evolution of the Computer

- First Generation
- Second Generation
- Third Generation
- Fourth Generation
- Fifth Generation

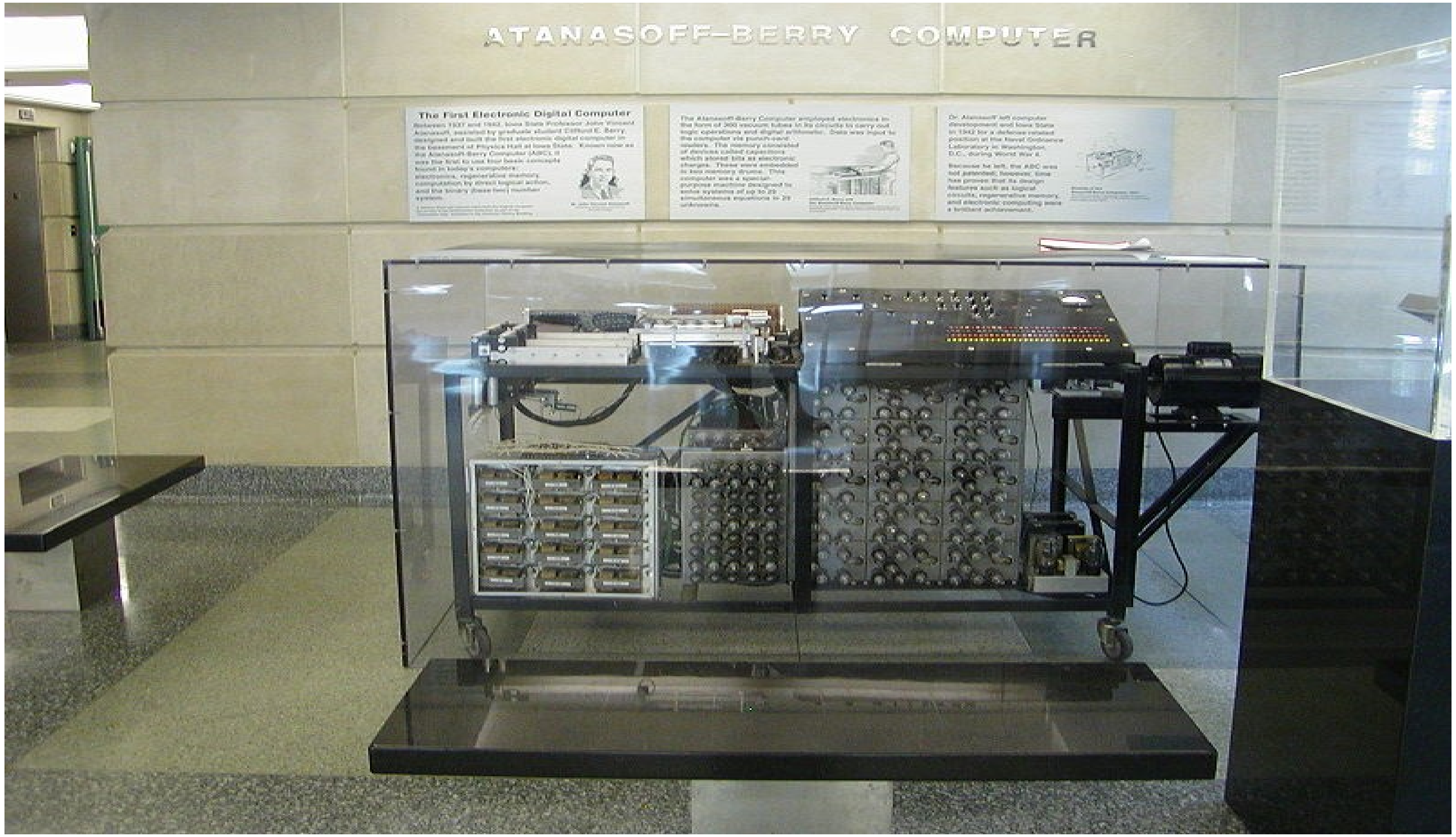
First Generation of Modern Computers (1941-1955)

Date	Scientist	Country	Name	Programmable	Binary	Electronic
May 1941	Conrad Zuse	Germany	Z3	Punched film	Yes	No
Summer 1941	John Atanasoff	Germany/ USA	ABC	No	Yes	Yes
1943	Tommy Flowers	UK	Colossus	By rewiring	Yes	Yes
1944	Howard Aiken/IBM	USA	Harvard Mark I	Punched paper tape	No	Yes
1944	Mauchly & Eckert	USA	ENIAC	By rewiring	No	Yes

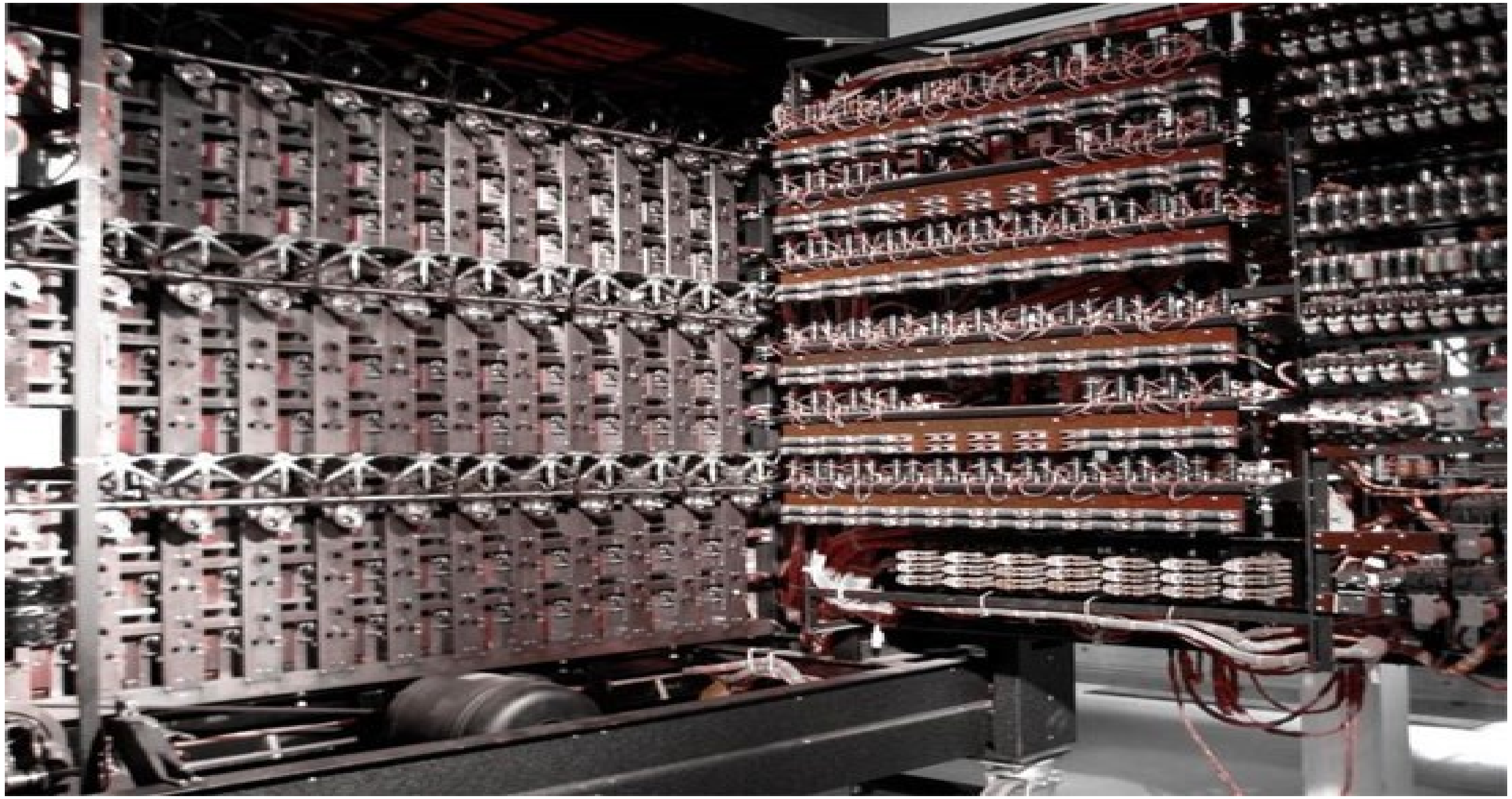
Zuse Z3 in Deutschen Museum in München



Atanasoff–Berry Computer replica at 1st floor of Durham Center, Iowa State University

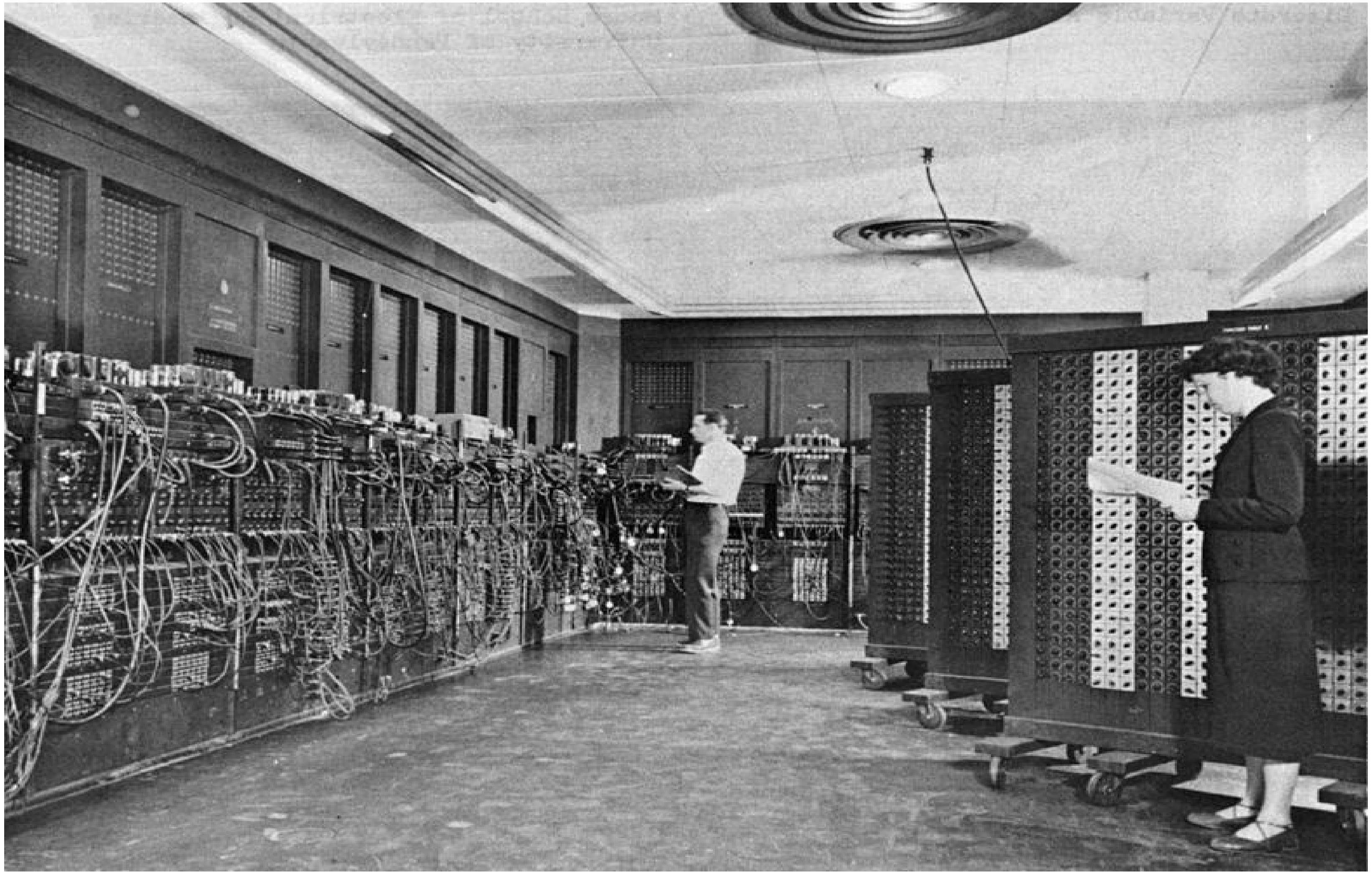


Colossus



Harvard Mark I

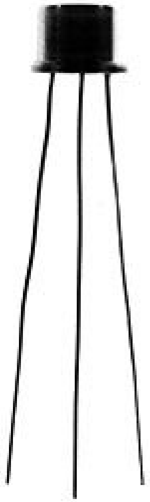
ENIAC



Second-Generation Computers (1956-1963)

- Transistor (1947) replaced vacuum tubes
  - Smaller and less power hungry: larger computers could be built

Vacuum Tube & Transistor



Third Generation Computers (1964-1971)

- Integrated Chips
  - (Many transistors integrated on the same chip)
  - Lower Power requirement
  - Faster Computers
- Software
  - Development of high level programming language
  - Easier to program computers
- Machines of this era started making inroads to the business enterprises
- Millions of operations per second

Fourth Generation Computers (1971-1995)

- Very Large Scale Integrated (VLSI) Chips
- Microprocessor
  - Lower Cost
  - Introduction of Personal Computers
- Software
  - Application software, Word processors, gaming etc.
  - Artificial Intelligence
- Embedded Systems and Robots
- Hundred million of operations per second or more

Fifth Generation Computers (1995-present)

- Parallel, Grid / Distributed Computing
- Mobile Computing
- Internet/World Wide Web
- Artificial Intelligence
- Software
  - Distributed Object Technologies
  - Middleware
    - Applications are constructed in layers or tiers
    - Maybe distributed along network

Future Trends

- Grid Computing Systems
- Enhancement in Personal/Mobile computing power
- Quantum Computing
  - Using sub-atomic particles (Neuron, Proton, Electrons) to store and manipulate data

Any Questions ?